
Internship Report

Many-body Scars in Rydberg Quantum Simulators

Doğan Özhan

Institute of Physics
Institut de Physique Nucléaire, Atomique et de Spectroscopie (IPNAS)
ULiège
Liège, Belgium

Lecture SSTG0016-1
Academic Year 2024-2025

Supervisor:

Peter Schlagheck

0 Acknowledgments

The author wishes to convey his sincere appreciation to his internship supervisor, Professor Peter Schlagheck, with whom the interviews were very enriching for the completion of this work, and helpful. The author also wants to thank him for his many explanations about the subject and interrelated topics of the internship.

The author is extremely grateful for his Professor Thierry Bastin who taught him the entirely bases of quantum mechanics, without which the completion of this work would have been impossible.

Special thanks are due to the author's girlfriend, who dedicated her time for proofreading and showed great kindness.

Finally, the author would like to thank his mother for her patience and versatility while the work was being written.

Contents

0	Acknowledgments	2
1	Introduction	4
2	Experimental and Theoretical Background	5
2.1	Experiment Framework / Description	5
2.1.1	Rydberg Simulator	5
2.1.2	Laser Excitation	5
2.2	Theoretical Aspect	6
2.2.1	Ergodicity Hypothesis	6
2.2.2	Eigenstate Thermalization Hypothesis	7
2.2.3	Mathematical Model	7
2.2.4	Example	8
3	Methodology	9
3.1	Hamiltonian Construction	9
3.2	Propagation Method for Schrödinger Equation	10
3.3	Measurements	11
3.3.1	z-Component of Spin	11
3.3.2	Correlation	14
4	Discussion	16
4.1	z-Component of Spin	16
4.2	Correlation	17
5	Conclusion	18

1 Introduction

Classical computers already won the heart of humans for almost century, with the conception of ENIAC (Electronic Numerical Integrator And Computer) in 1943 [1]. However, even though these latter are still useful nowadays, they are limited in different fields of application, which effectively slows the research in different fields of science in general. As a typical example, let us cite the well-known quantum cryptography problem of Shor’s algorithm [2],[3], which deals with factoring large numbers; in pair with a quantum computer, such an algorithm could break encryption systems like RSA in a finite time.

In addition, the arrival of quantum computers would make some advancements in surprising fields such as medical science [4]. Effectively, with those high-performance computers, one could simulate complex molecules that for example, can be found in biochemical field. As the dimension of the Hilbert space grows exponentially with the size of molecules, it is an arduous task, without saying impossible, for classical computers to simulate such grand molecules. While quantum computers are expected to simulate such complex systems, some prototype quantum computers already succeeded in simulating small molecules [5], namely in 2016, a dihydrogen molecule has been simulated [6], which is the simplest molecule that one can ever find.¹.

But cryptography and medical sciences are not the only potential fields of application for quantum computers: machine learning with the celebrated perceptrons for example [8]-[10], resolving laborious problems of optimization [11],[12], or even else in finance [13]-[15] and so more.

Yet, with the current knowledge and scientific advancement, it is not already possible to construct powerful enough quantum computers to simulate more complex systems like the one stated before, or to perform some machine learning problems quantumly. One day, in order to manage to construct a viable quantum computer, numerous researches on different fields have to be conducted, namely on many-body Scars in Rydberg quantum simulator.

This work specifically focuses on this current latter research subject, but also addressing the weak ergodicity breaking, as Scars and this latter phenomenon are deeply interrelated. Throughout this report, relevant definitions are provided, and the necessary theoretical background is reviewed in detail. In the following, all the numerical methodology that was followed is exposed as well as the results. The study is centered for the specific cases of 4, 5, 6, 7 and 8 atoms in the Rydberg simulator. It will be seen that the results suggest that many-body scars systems could engender new universality class of quantum dynamics, opening several opportunities for the future [16].

Ultimately, the purpose of this work, as it has been implied, is to enhance one’s knowledge to contribute to quantum computer scientific advancement and to pave the way for further research into these complex systems that are many-body scars.

¹The interested reader can also read this work on the limits of quantum computers simulations [7]

2 Experimental and Theoretical Background

Before delving into the core of the subject, some definitions/explanations are necessary to establish the framework of the experiment and to understand it. In this section, the experimental setup is described and theoretical background is provided as well in order to understand the study that has been carried out.

2.1 Experiment Framework / Description

It is relevant to mention that following explanations are exhibited for a material quantum simulator for more concreteness, however the approach that has been led is for a numerical quantum simulator.

2.1.1 Rydberg Simulator

Let us define a system constituted of a chain of atoms (for concreteness, let us imagine 50 atoms for now, all belonging to the same straight line, and it will be evident soon that the associated Hilbert space is then of dimension 2^{50}) where some of them are in a high excited state $|r\rangle$ (typically $|r\rangle = |70S_{1/2}\rangle$), sufficiently close to each other to interact when they are first neighbours, the interaction with second neighbours can with certainty be neglected, that is what is called a *Rydberg simulator*. If all the atoms were in their ground state $|g\rangle$, the system, supposed to be closed, is in equilibrium and thus will not evolve in time. Nonetheless, as in the actual case some of the atoms that constitute the system are excited, hence it is clear that this latter will evolve in time. The goal is to study how this physical system will evolve in time, and more particularly how some characteristics of the system will evolve in time ².

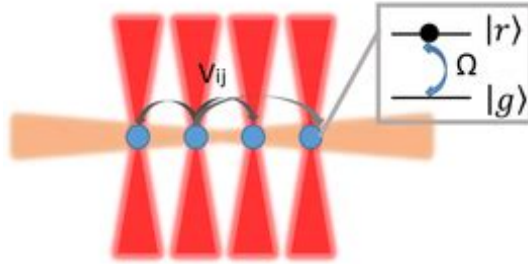


Figure 1: Depiction of a Rydberg quantum simulator. $|g\rangle$ denotes the ground state while $|r\rangle$ defines the Rydberg excited state. Ω is the transition pulsation between these two states and V_{ij} is the potential reigning between two neighboring atoms [18].

2.1.2 Laser Excitation

To excite some of the atoms, a laser of pulsation Ω is used (Ω trivially matches the transition energy between the two states $|g\rangle$ and $|r\rangle$ **if the involved atom is iso-**

²Some experimental details are skipped here, the interested reader can see the article "Probing many-body dynamics on a 51-atom quantum simulator" [17] for more detail

lated from any excited first neighbour). To prepare the system and make some experiments, several atoms are bombarded with the photons of the laser. When a non-excited atom becomes a *Rydberg atom*³, as the level of excitation is relatively high (i.e the distance between the excited electron and the nucleus, and the remaining electrons is relatively high), an electric dipole moment emerges between the excited electron of charge $-e$, and the nucleus together with the remaining electrons of effective charge $+e$, where $e = 1.602176634 \cdot 10^{-19} \text{ C}$ represents the elementary charge.

However, this electric dipole moment disrupts the system regarding its energy because an additional potential energy between the Rydberg atom and its two unexcited neighbours is created. Thus, the angular frequency ω required to excite one of its two neighbours is such that $\omega > \Omega$ ⁴, accordingly it is not absurd to consider that the laser at frequency $\frac{\Omega}{2\pi}$ can no longer excite the two neighbours.

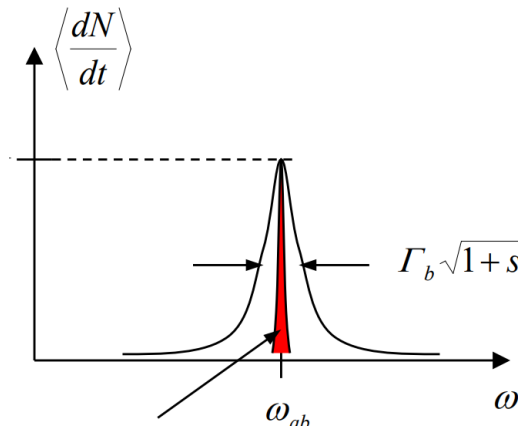


Figure 2: Lorentzian profile of the curve of the number of fluorescence cycles per second. ω_{ab} is the transition pulsation, Γ_b is the spontaneous emission rate and s is the saturation parameter [19].

2.2 Theoretical Aspect

In this section, the elementary principle of *ergodicity* is reviewed as well as the *eigenstate thermalization hypothesis* (ETH). In addition, the model that was used for the implementation of the experiment is given.

2.2.1 Ergodicity Hypothesis

In statistical physics, the *Ergodicity hypothesis* states that for all isolated systems in equilibrium, the time average of A is equal to the statistical average (or ensemble average) of A . In other words, for a given energy that restrains the phase space to a hypersurface $\Gamma(E)$, where E is the energy of the system, the representative point of the system in

³A *Rydberg atom* is an excited atom with one or more electrons that have a significantly high principal quantum number n [20].

⁴Even if it is not required that the laser has the exact same frequency of the transition because the population of excited atoms follows a lorentzian profile around the frequency of transition, we consider that the energy gap between the levels $|g\rangle$ and $|r\rangle$ is so large that the population that the laser at pulsation Ω can excite is totally negligible.

the restricted phase space will explore all the microscopic states that are found in this hypersurface over time (i.e all the microscopic states located in the hypersurface $\Gamma(E)$ are accessible and equiprobable. In equation, this reads

$$\rho(q_i, p_i) = \begin{cases} \text{constant,} & \text{if } (q_i, p_i) \in \Gamma(E) \\ 0, & \text{elsewhere} \end{cases}, \quad (1)$$

where $\{q_i\}$ and $\{p_i\}$ represent the ensemble of generalized coordinates and the associated generalized momenta respectively, and ρ can be interpreted as a probability density. As a consequence of this principle, as stated before, one could derive from it:

$$\overline{A_T} = \lim_{T \rightarrow +\infty} \frac{1}{T} \int_0^T A(t) dt = \int_{\Gamma(E)} A(\{q_i\}, \{p_i\}) \rho(\{q_i\}, \{p_i\}) dp_i dq_i = \langle A \rangle_T, \quad (2)$$

where it can be seen that the time average of A is equal to the statistical average of A [21].

2.2.2 Eigenstate Thermalization Hypothesis

The behaviour of quantum ergodic systems is governed by the *eigenstate thermalization hypothesis* [22],[23]. This latter states that individual eigenstates of quantum ergodic systems act as thermal ensembles, and thus the system's relaxation does not depend strongly on the initial conditions [24]. As an example of integrable system, the classical intuitive example of a clock pendulum can be mentioned: whatever the initial conditions are and without any additional perturbation, the pendulum will oscillate until it reaches its equilibrium state. It is said that the pendulum reaches its thermal equilibrium.

Now that these two main notions have been reminded, the case of many-body scars quantum systems can be tackled. Effectively, even though the emphasis on ergodic systems has been put, scars in quantum mechanics are special states that actually constitute non-ergodic systems. Moreover, relaxation of scars in Rydberg simulator strongly depends on the initial state of the system, some of these initial states give birth to a fast relaxation to thermal equilibrium, while others, as it will be seen soon, show periodic revivals before thermalization [16].

2.2.3 Mathematical Model

In the experiment that will be exposed, as implied, a chain of Rydberg atoms has been simulated, in which excitations were created and annihilated. The system is modelled by the following spin-1/2 Hamiltonian [16]

$$\hat{H} = \sum_{i=1}^L \hat{P}_i \hat{X}_{i+1} \hat{P}_{i+2}, \quad (3)$$

where $\hat{X}_j$, $\hat{Y}_j$ and $\hat{Z}_j$ represent the Pauli operators acting on the j^{th} atom, whose the matrix representation in the standard basis is given by

$$\hat{X}_j = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \hat{Y}_j = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \hat{Z}_j = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \forall j \in \{1, \dots, L\}, \quad (4)$$

and where L denotes the number of atoms that are in the Rydberg simulator (i.e the length of the chain). All the computations have been done in units $\hbar = 1$. In the following, the ground state and the excited Rydberg state are represented by the following kets $|0\rangle$ and $|1\rangle$ respectively. Knowing this, the Pauli operators can be written in Dirac Notation as follows

$$\hat{X}_j = |0\rangle\langle 1| + |1\rangle\langle 0|, \quad \hat{Y}_j = i(|1\rangle\langle 0| - |0\rangle\langle 1|), \quad \hat{Z}_j = |1\rangle\langle 1| - |0\rangle\langle 0| \quad \forall j, \quad (5)$$

where it can clearly be seen that the operator X_j acting on the j^{th} atom creates or annihilates an excitation on the site in question. Moreover, having recalled the Dirac notation for operators, the expression in terms of this latter of P_j appearing in equation (3) can be given as the common matrix notation

$$\hat{P}_j = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \equiv |0\rangle\langle 0| = \frac{1 - \hat{Z}_j}{2}, \quad (6)$$

which ensures that two successive atoms are not in a Rydberg state [16].

2.2.4 Example

The most important idea of the previous subsection to understand is the action of $\hat{P}_j \hat{X}_{j+1} \hat{P}_{j+2}$ on a group of 3 atoms. In order to grasp this concept, let us consider the j^{th} , $(j+1)^{th}$ and $(j+2)^{th}$ atoms. If the state of this three-body system is $|0, 0, 0\rangle$ initially, when $\hat{P}_j \hat{X}_{j+1} \hat{P}_{j+2}$ acts on it, the state $|0, 1, 0\rangle$ is obtained. Inversely, if the system is initially in the state $|0, 1, 0\rangle$, it gives $|0, 0, 0\rangle$. Finally, it can be said that the configurations $|1, 0, 0\rangle$, $|1, 0, 1\rangle$ and $|0, 0, 1\rangle$ are annihilated (i.e gives $|0, 0, 0\rangle$) under the action of the operator in question.

3 Methodology

In this section, the main lines of the python code's implementation is exposed. The method to obtain the Hamiltonian of the system is given as well as the experiments that have been carried out through this work, namely the measure of the z spin component for a given atom and initial state, and the correlation of spins between two neighbour atoms in the chain, also given an initial state. It is relevant to say that the atom's configuration was represented by lists of length L , while ket vectors were represented by lists of dimension $2^L - \# \text{ forbidden configurations}$.

3.1 Hamiltonian Construction

In order to know how a unitary system will evolve through time according to the Schrödinger equation⁵ in quantum mechanics, it is necessary to possess the Hamiltonian of this system. In Subsection 2.1.2, it is said that two following atoms cannot be both in a Rydberg state. From this information, it can be derived that if an atom is excited (i.e is in state $|1\rangle$), it inevitably has two non-excited neighbours (i.e two neighbours in state $|0\rangle$), otherwise the system's configuration would be impossible. This consideration trivially constrains the Hilbert space of dimension 2^L ⁶. Hence, system states as $|\dots, 1, 1, \dots\rangle$ are formally prohibited. With the aim of not dealing with those "*dark states*" [16], it has been decided to work, through all the implementation of the code, with a restricted-dimension Hamiltonian⁷.

Once all this information has been provided, the procedure that has been followed to construct the Hamiltonian can be given. For all of the possible configurations of atoms, if there was a succession of three one-body state $|0\rangle$ in the state representation of the system, then the "0" in the middle was replaced by a "1". The same rule has been followed for the atoms on the edges: if its only neighbour is in state $|0\rangle$ and if it is also in the same state, then the $|0\rangle$ state on the edge was turned into the $|1\rangle$ state. Moreover, by swapping the roles of one-body states $|0\rangle$ and $|1\rangle$ in the two previous sentences, a "0" was placed where there was a "1" in the relevant possible configuration. In this way, for example, if $L = 6$ and one had to deal with the configuration portrayed by $[0, 1, 0, 1, 0, 1]$, the state number 21⁸ the three configurations: $[0, 0, 0, 1, 0, 1]$, $[0, 1, 0, 0, 0, 1]$, and $[0, 1, 0, 1, 0, 0]$ (state number 5, 17 and 20) would have been obtained. Thus, in the Hamiltonian of dimension 21, three numbers "1" were being placed in the positions relative to these 3 only potential configurations, and so eighteen numbers "0" were placed in the row corresponding to the state number 21 as well, for the other states (for example, for configurations $[0, 0, 0, 0, 0, 1]$, $[1, 0, 0, 0, 0, 1]$, or else for configurations $[0, 0, 1, 0, 0, 0]$ and $[0, 1, 0, 0, 0, 0]$). In other words, a "binary Hamiltonian" with a high number of "0" has been constructed.

⁵Due to Erwin Rudolf Josef Alexander Schrödinger, an Austrian physicist, (born in Vienna the 12 August 1887, dead the 4 January 1961) who was awarded the Nobel Prize in 1933 for this celebrated equation [25].

⁶Let us remind that the number of atoms is denoted by the letter L . The non-constrained space is of dimension 2^L because all the atoms constituting the N -body system can be in only two different states, $|0\rangle$ or $|1\rangle$, irrespectively of each other and trivially because there are L atoms.

⁷The interested reader can verify that if there are 6 atoms, instead of dealing with a Hamiltonian of dimension 64, one would have to deal with a 21-dimension Hamiltonian.

⁸The sharp-eyed reader would have noticed that the decimal numbering of the list is given according to the binary number that presents the list.

For concreteness, here is the Hamiltonian that was constructed for a number of atoms $L = 4$

$$\hat{H} = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \end{bmatrix},$$

where each row trivially corresponds to a numbered state, such that if row i corresponds to state number n , and row j to state number n' , then $i < j$ implies $n < n'$. In other words, the vectors that constitute the basis in which the Hamiltonian is represented are "classified in ascending order" (in binary representation).

3.2 Propagation Method for Schrödinger Equation

Once the Hamiltonian has been obtained, an estimation of the wave function can be computed through a simple algorithm that is explained in this subsection.

The evolution of the system that is considered here is driven by the famous Schrödinger equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle, \quad (7)$$

or else in $|\mathbf{r}\rangle$ representation

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{r}, t) = \hat{H}(t) \psi(\mathbf{r}, t).$$

Yet, in the current case, the Hamiltonian does not depend on time, thus equation (7) is reduced to a constant-coefficient differential equation, which is substantially simple to solve

$$\begin{aligned} i\hbar \frac{d}{dt} |\psi(t)\rangle &= \hat{H} |\psi(t)\rangle \\ \Leftrightarrow |\psi(t)\rangle &= |\psi(0)\rangle e^{-\frac{i}{\hbar} \hat{H} t}, \end{aligned}$$

where $|\psi(0)\rangle$ is the state at the initial time $t = 0$. However in the experiment that has been led, $\hbar$ was set to 1 to get rid of physical unities. Thus, the latter equation in the current case can be rewritten as

$$|\psi(t)\rangle = |\psi(0)\rangle e^{-i\hat{H}t} . \quad (8)$$

In such a case, numerically, one could try to diagonalize the Hamiltonian in order to compute the exponential of such a matrix with, for example, the Jacobi algorithm [26]. Nevertheless, such diagonalization algorithms quickly become computationally expensive in practice, hence another approach has been followed through this work. The definition of the exponential was simply harnessed

$$e^x = \sum_{j=0}^{+\infty} \frac{x^j}{j!} ,$$

as trivially expected, the exponential in eq (8) was simply expressed in terms of its series definition and truncated at the M^{th} term (typically $M = 20$). As a result, this gives the approximation:

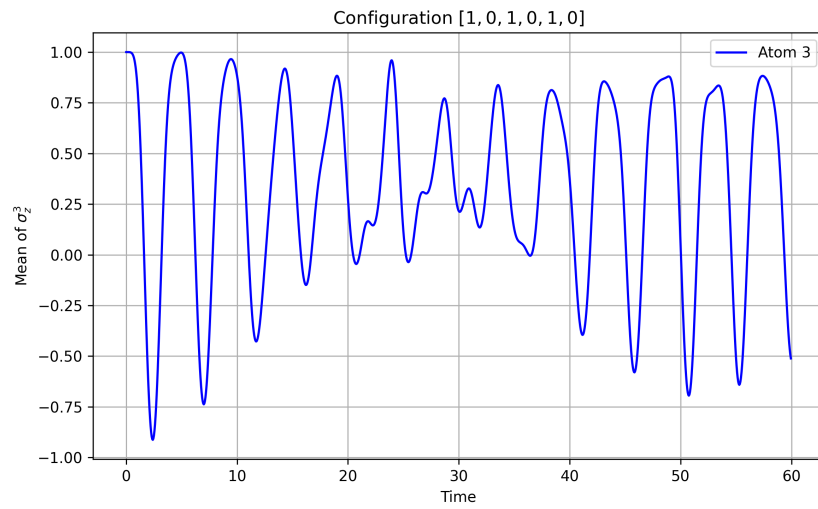
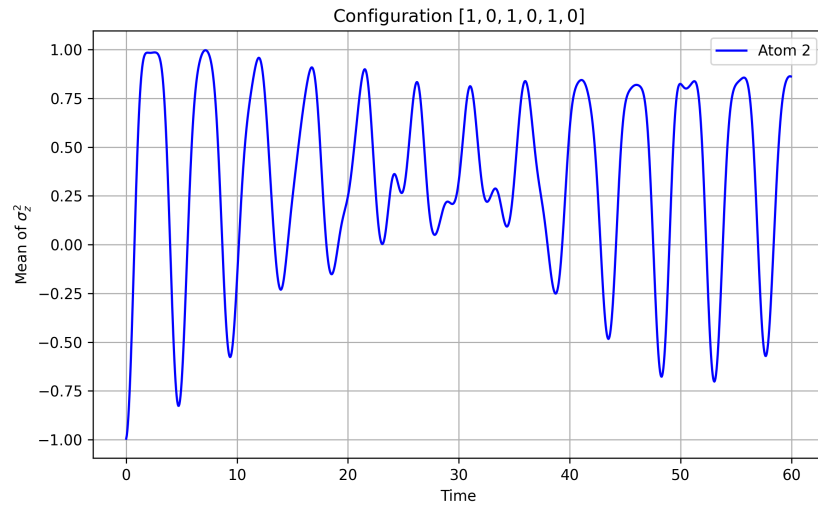
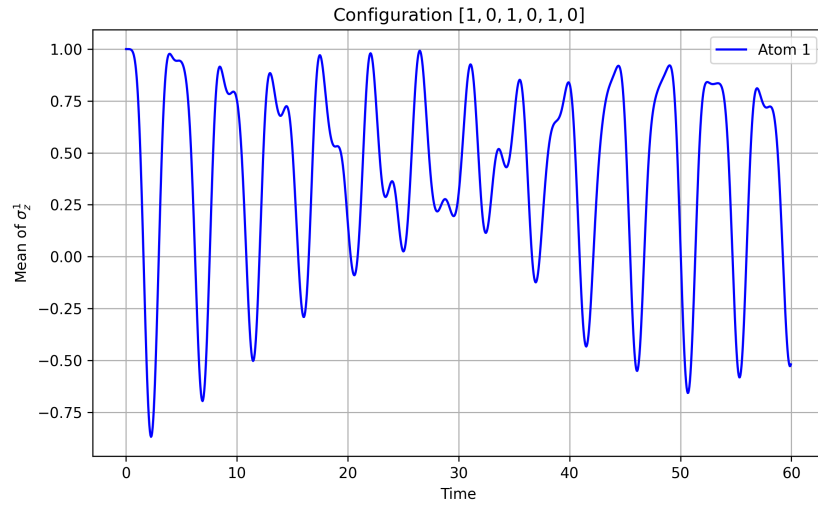
$$|\psi(t)\rangle \simeq |\psi(0)\rangle \sum_{j=0}^M \frac{(-it)^j}{j!} \hat{H}^j . \quad (9)$$

From a theoretical point of view, the wave function can clearly be computed for all t . Nonetheless, in practice and actually, what has been done in the current experiment, one would rather compute first $|\psi(0 + \delta t)\rangle$, then uses this new ket as the initial state to propagate and obtain $|\psi(\delta t + \delta t)\rangle$, and so on until reaching $|\psi(2\delta t + \delta t)\rangle$ and so on until obtaining $|\psi(n\delta t)\rangle \equiv |\psi(t)\rangle$. This step of computation is called the *propagation*.

3.3 Measurements

3.3.1 z-Component of Spin

As first characteristic of the system to be measured, it has been decided to choose the z component of spin as a function of time obviously. Here are plotted 4 graphics of this quantum quantity:



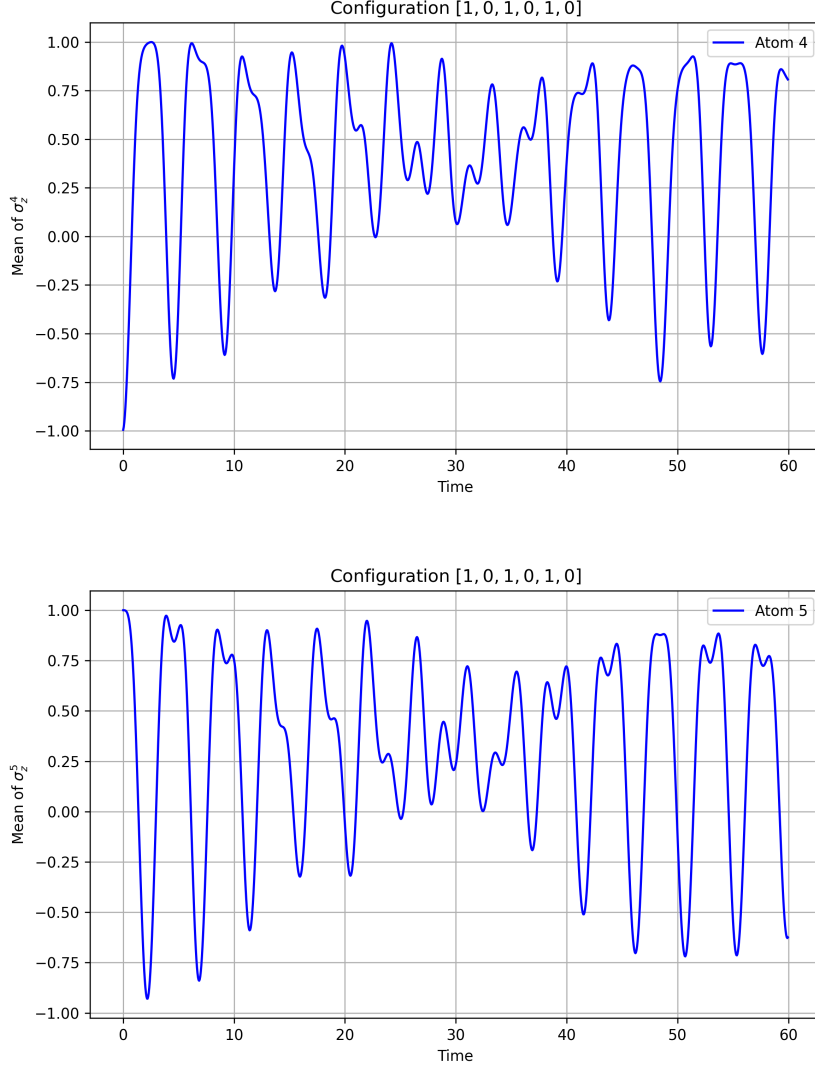


Figure 3: Mean of σ_z^f for different atoms that form the $L=6$ Rydberg chain with the initial configuration $[1, 0, 1, 0, 1, 0]$, where f represents the number of the atom in this latter chain. The atoms are numbered as $[0, 1, 2, 3, 4, 5]$, so the odd places are occupied by atoms in single initial state $|0\rangle$, and even places by atoms in single initial state $|1\rangle$.

To make it easier to compare numbered odd and even atoms, here is a superposition of two such graphics for the atoms numbered 1 and 2 for the initial configuration $[1, 0, 1, 0, 1, 0]$.

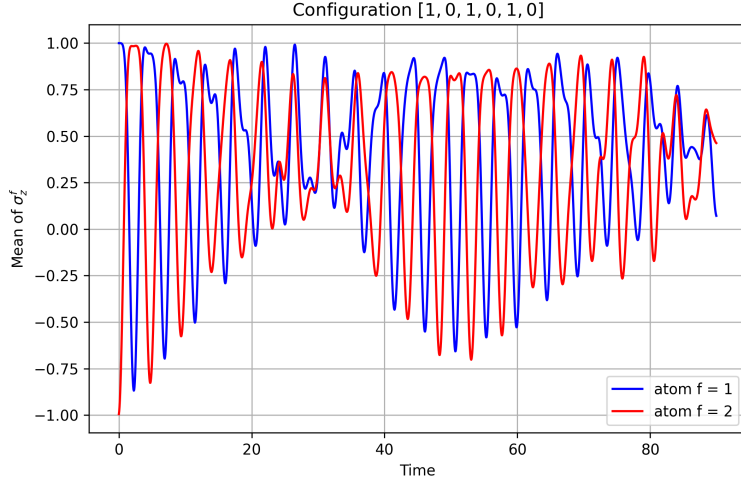
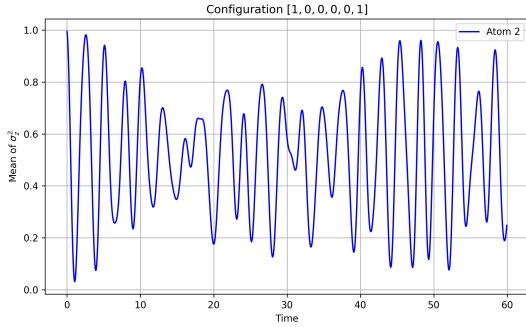
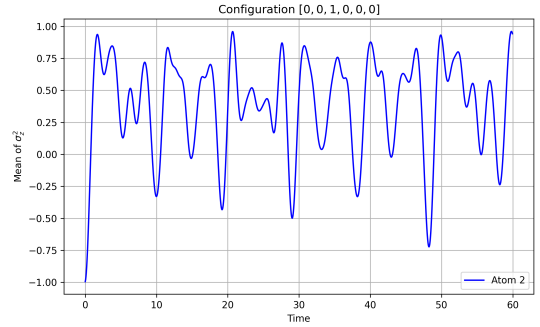


Figure 4: Superposition of two previous plots: corresponding to $f = 1$ and $f = 2$

Two initial configurations for $L = 6$ have been tested as well for the sake of pure curiosity.



(a) Configuration α



(b) Configuration β

Figure 5: Graphs of the time evolution of the mean of σ_z^2 over time⁹.

3.3.2 Correlation

The other and last characteristic that has been measured is the spin-spin correlation, which the mathematical expression is given by

$$\sum_{f=0}^{L-2} \langle \sigma_z^f \rangle \langle \sigma_z^{f+1} \rangle . \quad (10)$$

The corresponding graphs are shown below.

⁹Careful! The "2" in exponent here represents the number of the atom in the chain, and not the square of the operator σ_z .

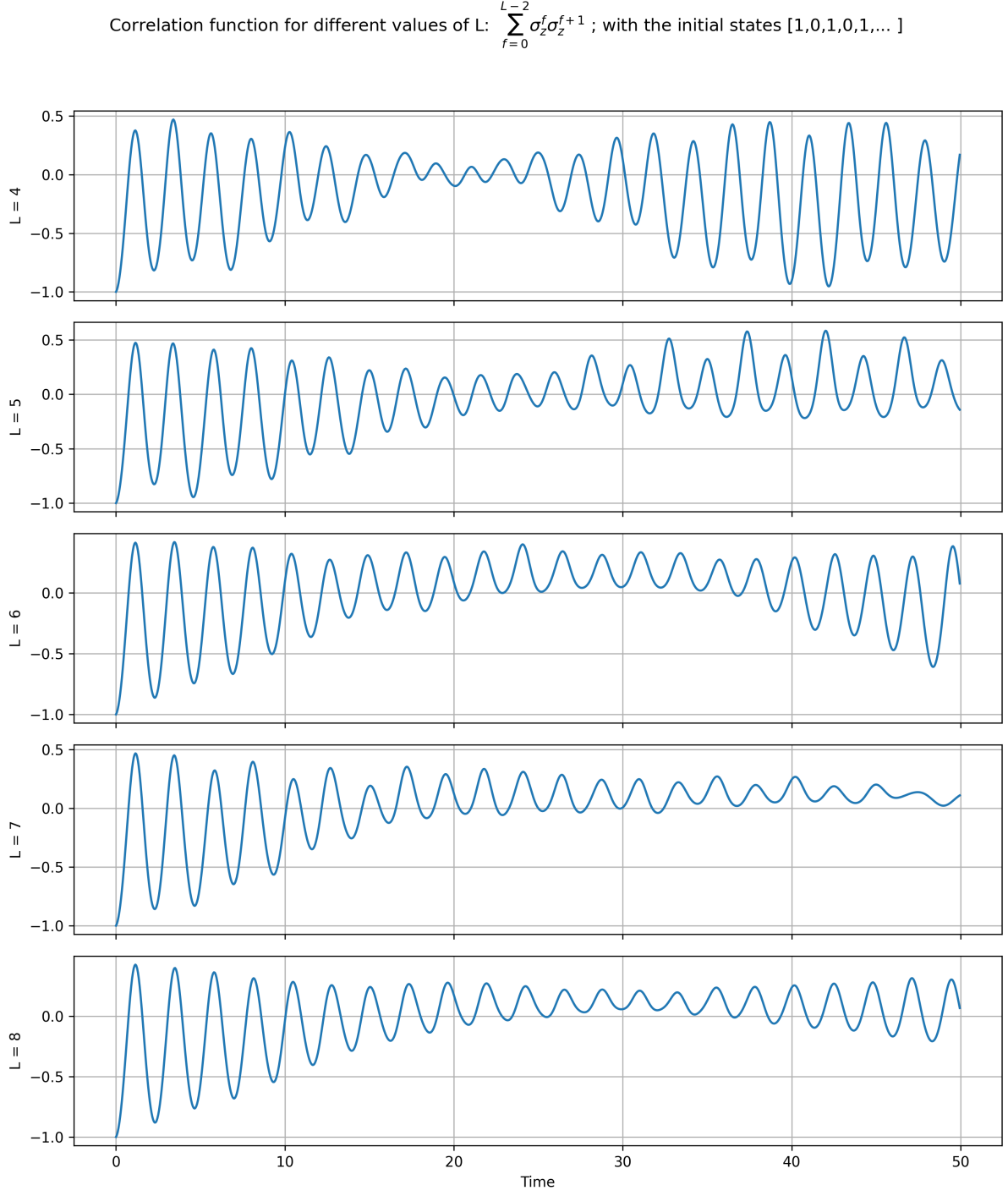


Figure 6: Correlation between z component of spin for $L = 4, 5, 6, 7$ and 8 , for the initial states $|\mathbb{Z}_2\rangle$ ¹⁰

¹⁰Period-2 charge density wave states [16] $|\mathbb{Z}_2\rangle$ are defined as $|r, g, r, g, r, \dots\rangle$, while in generality $|\mathbb{Z}_k\rangle$ are defined as $|\dots, r, g, \dots, g, r, \dots\rangle$ where there are $k - 1$ atoms in single atom state $|g\rangle$ between the two states $|r\rangle$ displayed.

4 Discussion

In this section, a brief analysis of the graphics displayed in the previous section is carried out. Effectively, the noteworthy details will be discussed in order to identify some specificities of such quantum scars systems.

4.1 z-Component of Spin

Let us first discuss the measurements of z-components of spin. For the configuration $|\mathbb{Z}_2\rangle$ ($L = 6$), it can first be stated that the z-component of spin of each atom that constitutes the chain, oscillates with an almost-constant period. This latter might not be constant due to the numerical computation, and also probably due to the fact that the exponential series that has been used to compute the state of the system has been truncated, as explained in Section 3.2. In addition, beyond this apparent period, one might wonder if there is also another periodic time scale in the system, this is what has been investigated.

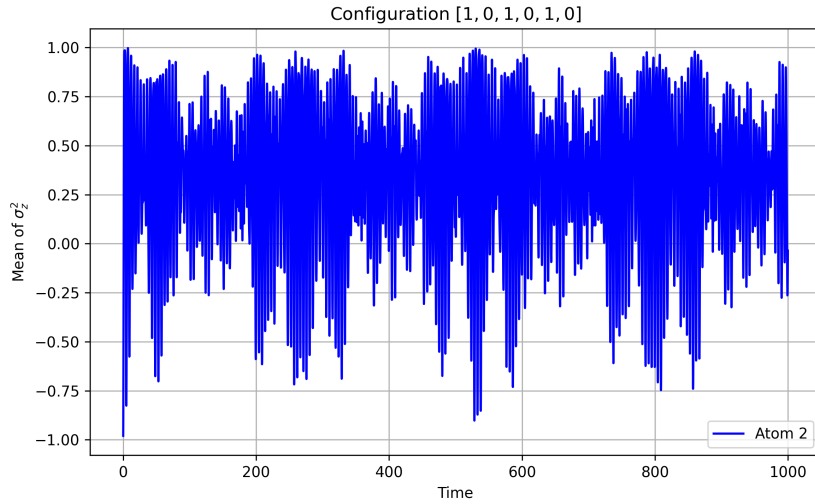


Figure 7: Evolution of $\langle \sigma_z^2 \rangle$ according to time. 10000 propagation steps have been performed in order to obtain this graph¹¹.

It can clearly be seen that the system presents another larger period, there is always a succession of 3 crude peaks in the revivals of the mean value of σ_z^2 ¹². This behavior is obviously the same for each even atom in the $|\mathbb{Z}_2\rangle$ configuration, while that of the odd atoms is exactly the same, but with a phase shift, as it can be noticed in the graphics exposed in Section 3.3.

Finally, concerning the mean σ_z measurement, based on the graphs corresponding to the configurations previously named α and β , one can state that the evolution of the Rydberg one-dimensional lattice depends on the initial state of the system. Moreover, a periodic behaviour is harder to identify in these two previously cited cases, nothing can therefore be concluded with these two pieces of information alone.

¹¹Same remark as the penultimate footnote.

¹²Idem.

4.2 Correlation

Regarding experiments conducted for correlation measurement, it can be stated, for the $|\mathbb{Z}_2\rangle$ configuration, that the correlation, as expected, oscillates. It means that over time, each atom alternates between spin up and spin down. In addition, the system appears to thermalize more slowly as the value of L increases. Moreover, it seems that the correlation between spins never exceeds the value of 0.5 and this holds regardless of the number L of atoms. This latter result reflects the fact that the atoms in the Rydberg chains have never, at each time, their spin aligned in the same direction. Finally, as briefly discussed in Section 2, this is now evident that the system shows periodic revivals, instead of just thermalize.

5 Conclusion

In conclusion, this study provided obvious evidence to the fact that quantum many-body scars are "special" states that weakly break ergodicity [16]. Through thorough computer simulations, this particular behaviour was made evident. Different configurations have been investigated, changing the number of atoms that compose the Rydberg chain and the initial configuration of the system as well.

The description of the experimental setup was crucial in order to have a concrete representation of the phenomenon that was under study. Effectively, a brief reminder of laser excitation was provided, which clearly posed the framework. Moreover, some theoretical background was discussed as well: the ergodicity hypothesis, which was central in the current work of quantum scars; and the celebrated eigenstate thermalization hypothesis as well, which was as important as the previously stated hypothesis in order to understand the results.

The PXP model has been used to implement the system's dynamics, taking into account that two neighbour atoms cannot be both in the Rydberg state. This model has been implemented in lines of code, which gave rise to, what was called, a binary Hamiltonian.

Concerning the methodology, a necessary method to compute the state of the quantum system via the Schrödinger equation was given. Thanks to this method, the propagation was successful and allowed one to compute some characteristics of the system, namely the mean of the z component of spin of the atom as well as the correlation between neighboring atoms's mean spin z -components between neighbour atoms.

The outcome of this study provided promising results to future studies for quantum many body scars and weak ergodicity breaking. Even though only a small number of atoms was used in the simulations, the notable behaviour of such systems was successfully observed. Effectively, the periodic revivals were clearly evident to identify. It can be added that, future studies could be based on such elementary research to build other theories/experiments in this field.

Finally, it must be acknowledged that this study still lacks certain information, leaving the field open to complementary and future research. Mastering the dynamics that governs quantum many body scars would represent a major step forward in quantum computing in order to control quantum coherence of states.

References

- [1] Michael T. Swaine, Paul A. Freiberger, "ENIAC", *Britannica* (Jan 25, 2025). Available at: <https://www.britannica.com/technology/ENIAC> (consulted March 16, 2025 at 16:13).
- [2] QuEra, "Shor's Algorithm", *QuEra* (2025). Available at: <https://www.quera.com/glossary/shors-algorithm> (consulted March 16, 2025 at 17:00).
- [3] F. Damanet, "PHYS0997-1: Quantum information and computation", *Université de Liège* (2024-2025).
- [4] S.Alrashed, N. Min-Allah, "Quantum computing research in medical sciences", *Informatics in Medicine Unlocked*, Vol.52 (2025). Available at: <https://www.sciencedirect.com/science/article/pii/S2352914824001631> (consulted March 17, 2025 at 17:30).
- [5] A.S. Kesari, S. K. Chaganti, C. Chowdhury, "Quantum simulation of small molecules for multiple applications", *Chemical Physics*, Vol.580 (April 1, 2024). Available at: <https://www.sciencedirect.com/science/article/pii/S0301010424000247> (consulted March 17, 2025 at 18:12).
- [6] P. Ball, "Quantum computer simulates hydrogen molecule", *ChemistryWorld* (July 25, 2016). Available at: <https://www.chemistryworld.com/news/quantum-computer-simulates-hydrogen-molecule-/1010041.article> (consulted March, 17 at 18:05).
- [7] Y. Zhou, E. M. Stoudenmire, X. Waintal, "What Limits the Simulation of Quantum Computers?", *Physical Review Journals* (November 23, 2020). Available at: <https://journals.aps.org/prx/abstract/10.1103/PhysRevX.10.041038> (consulted March 17, 2025 at 19:40).
- [8] A. J. da Silva, T. B. Ludermir, W. R. de Oliveira, "Quantum perceptron over a field and neural network architecture selection in a quantum computer", *Neural Networks*, Vol.76 (April 2016). Available at: <https://www.sciencedirect.com/science/article/pii/S0893608016000034> (consulted March 18, 2025 at 11:52).
- [9] M. Schuld, I. Sinayskiy, F. Petruccione, "Simulating a perceptron on a quantum computer", *Physics Letters A*, Vol.379, Issue 7 (March 2015). Available at: <https://www.sciencedirect.com/science/article/pii/S037596011401278X> (consulted March 18, 2025 at 11:55).
- [10] L. Lyu, C. Pang, J. Wang, "A quantum model of biological neurons", *Neurocomputing*, Vol.598 (September 2024). Available at: <https://www.sciencedirect.com/science/article/pii/S0925231224009949> (consulted March 18, 2025 at 11:57).
- [11] L. Gyongyosi, "Approximation Method for Optimization Problems in Gate-Model Quantum Computers", *Chaos, Solitons & Fractals: X*, Vol.7 (December 2021). Available at: <https://www.sciencedirect.com/science/article/pii/S2590054421000117> (consulted March 20, 2025 at 20:52).

- [12] R. M. Abdullah, R. Basher, A. A. Alwan, A. Z. Abualkishik, "Quantum Computers for Optimization the Performance", *Procedia Computer Science*, Vol.160, P54-60 (2019). Available at: <https://www.sciencedirect.com/science/article/pii/S1877050919316588> (consulted March 20, 2025 at 20:56).
- [13] R. Orús, S. Mugel, E. Lizaso, "Quantum computing for finance: Overview and prospects", *Reviews in Physics*, Vol.4 (November 2019). Available at: <https://www.sciencedirect.com/science/article/pii/S2405428318300571> (consulted March 20, 2025 at 21:00).
- [14] Y. Lu, J. Yang, "Quantum financing system: A survey on quantum algorithms, potential scenarios and open research issues", *Journal of Industrial Information Integration*, Vol.41 (September 2024). Available at <https://www.sciencedirect.com/science/article/pii/S2452414X24001079> (consulted March 20, 2025 at 21:02).
- [15] D. Zhai, T. Zhang, G. Liang, B. Liu, "Quantum carbon finance: Carbon emission rights option pricing and investment decision", *Energy Economics*, Vol.134 (June 2024). Available at: <https://www.sciencedirect.com/science/article/pii/S0140988324003360> (consulted March 20, 2025 at 21:05).
- [16] C. J. Turner, A. A. Michailidis, D. A. Abanin, M. Serbyn and Z. Papić, "Weak ergodicity breaking from quantum many-body scars", *nature physics* (May 2018). Available at: <https://www.nature.com/articles/s41567-018-0137-5> (consulted March 21, 2025 at 12:19).
- [17] H. Bernien, S. Schwartz, A. Keesling, H. Levine, A. Omran, H. Pichler, S. Choi, A. S. Zibrov, M. Endres, M. Greiner, V. V. & M. D. Lukin, "Probing many-body dynamics on a 51-atom quantum simulator", *nature physics* (November 2017), vol. 551, p580. Available at: <https://www.nature.com/articles/nature24622> (consulted March 23, 2025 at 17:27).
- [18] M. Wu, "Rydberg-atom simulator and quantum many-body scars", *Department of Physics, Stanford University, Stanford, CA 94305* (June 27, 2020). Available at: <http://large.stanford.edu/courses/2020/ph470/wu/docs/wu-ph470-stanford-spring-2020.pdf> (consulted March 21, 2025 at 16:54).
- [19] T. Bastin, "PHYS0932-1: Atomes froids et horloges atomiques", *Université de Liège* (2024-2025).
- [20] Wikipedia, "Rydberg atom", *Wikipedia* (February 10, 2025). Available at https://en.wikipedia.org/wiki/Rydberg_atom (consulted March 23, 2025 at 12:05).
- [21] N. Vandewalle, "PHYS0212-2: Physique statistique", *Université de Liège* (2023-2024), p47.
- [22] J. M. Deutsch, *Quantum statistical mechanics in closed system*, Phys. Rev. A 43, 2046-2049 (1991)
- [23] M. Srednicki, *Chaos and quantum thermalization*, Phys. Rev. E 50, 888-901 (1994)

- [24] M. Serbyn, D. A. Abanin & Z. Papić, "Quantum many-body scars and weak breaking of ergodicity", *nature physics* vol. 17, pp. 675-685 (2021)
- [25] The Nobel Prize, "Erwin Schrödinger Biographical", *The Nobel Prize* (last update 2025). Available at <https://www.nobelprize.org/prizes/physics/1933/schrodinger/biographical/> (consulted March 30, 2025 at 12:11)
- [26] T. Bastin, "PHYS0209-3: Méthode numérique de la physique", *Université de Liège* (2022-2023).